

Exploring Microsoft Copilot & AI Agents to Transform Retail Operations

——Transforming Questionnaire Creation with Human-AI Collaboration

Presented to AS Watson & Microsoft | HKU Capstone Project

Group 1 — Questionnaire & Survey Creation Use Case



Team Member Introduction



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Name	Responsible work
Liu Mingshan	Coordinate project progress, set up an initial demo for the entire platform, and design the overall workflow
Li Yunxiao	Responsible for demo design of the platform homepage section
Ding Zijian	Responsible for the demo design of the platform questionnaire drafting module
Liao Shujun	Responsible for the demo design of the platform questionnaire processing module
Zhao jiale	Responsible for the demo design of the platform approval flow module.
Sun Keming	Responsible for the demo design of the platform questionnaire logic check module .
Wang Siming	Responsible for the demo design of the platform questionnaire translation module .
Huang Deyue	Market research, competitive product case analysis, summary of key conclusions, PPT production

Note: Arrange according to the display order

Project Background

This HKU capstone project, in collaboration with **Microsoft** and retail partner **AS Watson**, focuses on using Microsoft Copilot and AI agents to optimize the end-to-end questionnaire and survey creation workflow.



AS Watson Reality – Scale Creates Complexity

- AS Watson is a global health & beauty retailer operating across multiple markets and **5–10+ languages**.
- A single survey may contain **40–60 questions** with **hundreds of branching logic paths**.
- Surveys are a critical source of **Consumer Market Intelligence** (e.g., product testing, brand tracking, satisfaction).
- The current manual, multi-step process is a **bottleneck** to market agility.

1 Survey

5-10
Languages

40-60
Questions

100+
Paths



The Microsoft AI Opportunity

- M365 Copilot:** Enhances personal productivity within familiar apps (Excel drafting, summarization).
- Copilot Studio:** Enables building Custom AI Agents to automate multi-step business processes—the missing orchestration layer.
- Azure OpenAI & Foundry:** Provides the advanced reasoning required to parse complex logic trees and generate test cases.
- Power Automate:** Connects M365 apps and services into automated workflows, handling file movement, approvals, and cross-system triggers.



Project Mission:

The project aims to investigate how **Microsoft Copilot and AI Agents** — AI capabilities embedded in everyday productivity tools and extensible through Microsoft Copilot Studio — can deliver measurable operational and customer experience improvements across **retail business functions**.



Current Workflow – The 10-Step Process

Tools: Excel (Drafting & Design); 3rd-party survey platform (Build & Deployment); **Non-API enabled**

Key Teams: Market Intelligence Team (end-to-end process), Legal/Compliance, Category Teams, Regional Leads, DataLab and Local Business Units. (**already proficient in using LLMs for daily work**)

Current Process Breakdown

#	Step	Primary Activities	Time %	Category
1	Conception & Drafting	Design questions and define logic in Excel.	15%	Human Design
2	Design Review & Validation	Check Excel for duplicate codes/options, validate basic logical dependencies.	10%	Verification
3	Internal Review & Verification	Content and compliance review by Legal, Category, and Regional leads.	10%	Verification
4	Translation Coordination	Distribute finalized questionnaire to Business Units for localization.	10%	Translation
5	Translation Review & Alignment	Review multiple language versions for structural consistency and terminology accuracy.	3%	Translation
6	Survey Build & Configuration	Manually rebuild the questionnaire on the 3rd-party survey platform (e.g., Wise) based on Excel.	12%	Manual Build
7	Build Verification & Reconciliation	Export platform configuration and cross-check line-by-line against the original Excel.	15%	Verification
8	Test Scenario Generation	Generate end-to-end test cases covering all branching logic paths.	8%	Testing
9	End-to-End Testing	Simulate user journeys for all test scenarios to validate system behavior.	15%	Testing
10	Publication & Go-Live	Deploy survey on the platform, configure scheduling and audience.	2%	Deploy



Key Insight:

Over **60%** of total cycle time is consumed by manual data entry, verification, and testing — steps 6-9 alone account for **50%** of total effort. However, Not one step is the most “time-consuming” or repetitive. Productivity gain will come from **process re-engineering**.

Core Pain Points – Why “Linear” Breaks at Scale



Linear Effort Scaling

The Problem

Survey volume requires proportional headcount growth. The most fundamental scaling challenge is the **linear relationship** between survey volume and manual effort.

Impact at Scale

- Each new survey repeats the full 10-step manual process
- Headcount must grow proportionally with survey volume
- No economies of scale in the current workflow



Manual End-to-End Testing

The Problem

Test scenarios covering hundreds of branching logic paths are generated and validated manually. Human testers cannot exhaustively cover every possible paths.

Impact at Scale

- Consumes ~15% of total cycle time for each survey release
- Easy to overlook the edge-case logic errors
- Post-distribution fixes are costly — zero tolerance for flawed surveys at scale



Multi-Language Bottleneck

The Problem

5-10+ language versions create exponential synchronization complexity. Each language requires separate platform setup and cross-checks.

Impact at Scale

- Time costs scale linearly with language count
- Version misalignment risk increases exponentially
- Sequential translation blocks global release timelines



Platform Configuration Friction

The Problem

Survey logic is manually rebuilt in the platform because the import feature demands a strict, non-standard format that differs from the working Excel template.

Impact at Scale

- Manual rebuild is time-consuming and error-prone
- AI could automate formatting, but any conversion error triggers full manual re-verification — negating time savings



Objectives:

Process re-engineering



50–70% reduction in cycle time



No rise in error rate

Two Main Sources of Efficiency Improvement

1 Shift Repetitive Work to Machines

Core Transformation

- Move rule-based, highly repetitive operational tasks (~60%) out of manual workflows and **let AI handle them automatically.**

Value

- Manually upgrading from the “execution layer” to the “**decision-making layer**”, only requires final review, significantly reducing labor costs.

Repetitive Task	How to Change?
Survey Structure Parsing	Automatically parse questionnaire structure and logical branches
Logic Integrity Checking	Full-branch intelligent checking, zero missed errors
Test Case Generation & Execution	Automated full coverage testing answer path, doubling the efficiency of review
Multi-Language Translation	Automatic translation with consistency guaranteed

2 Eliminate Departmental Wait Times

Core Transformation

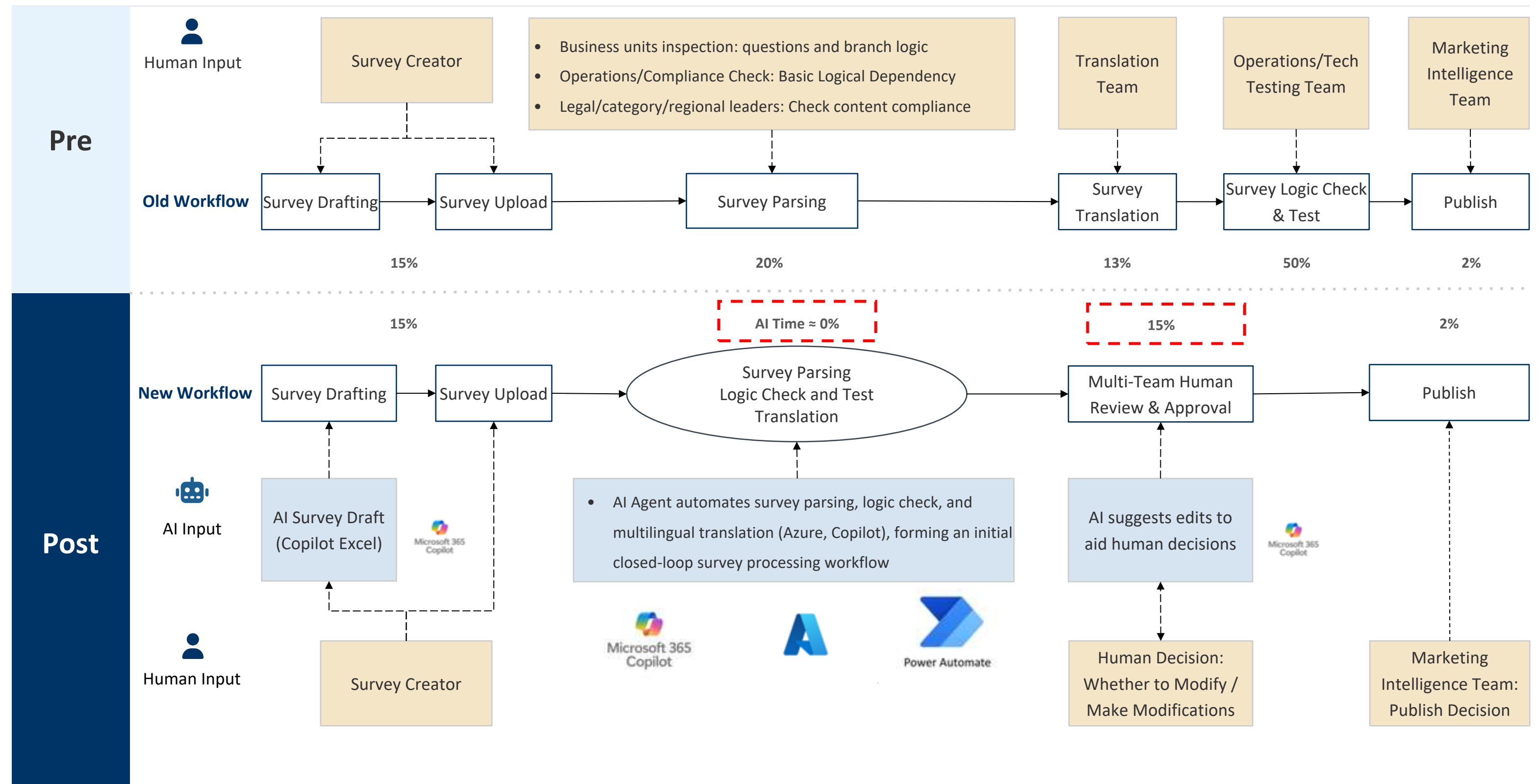
- The old workflow’s serial steps are restructured into a parallel, automated approval process — eliminating cross-departmental waiting costs.

The Mechanism: “Professional Serial Nodes + Localized Parallel Editing + Precision Impact Re-Review”

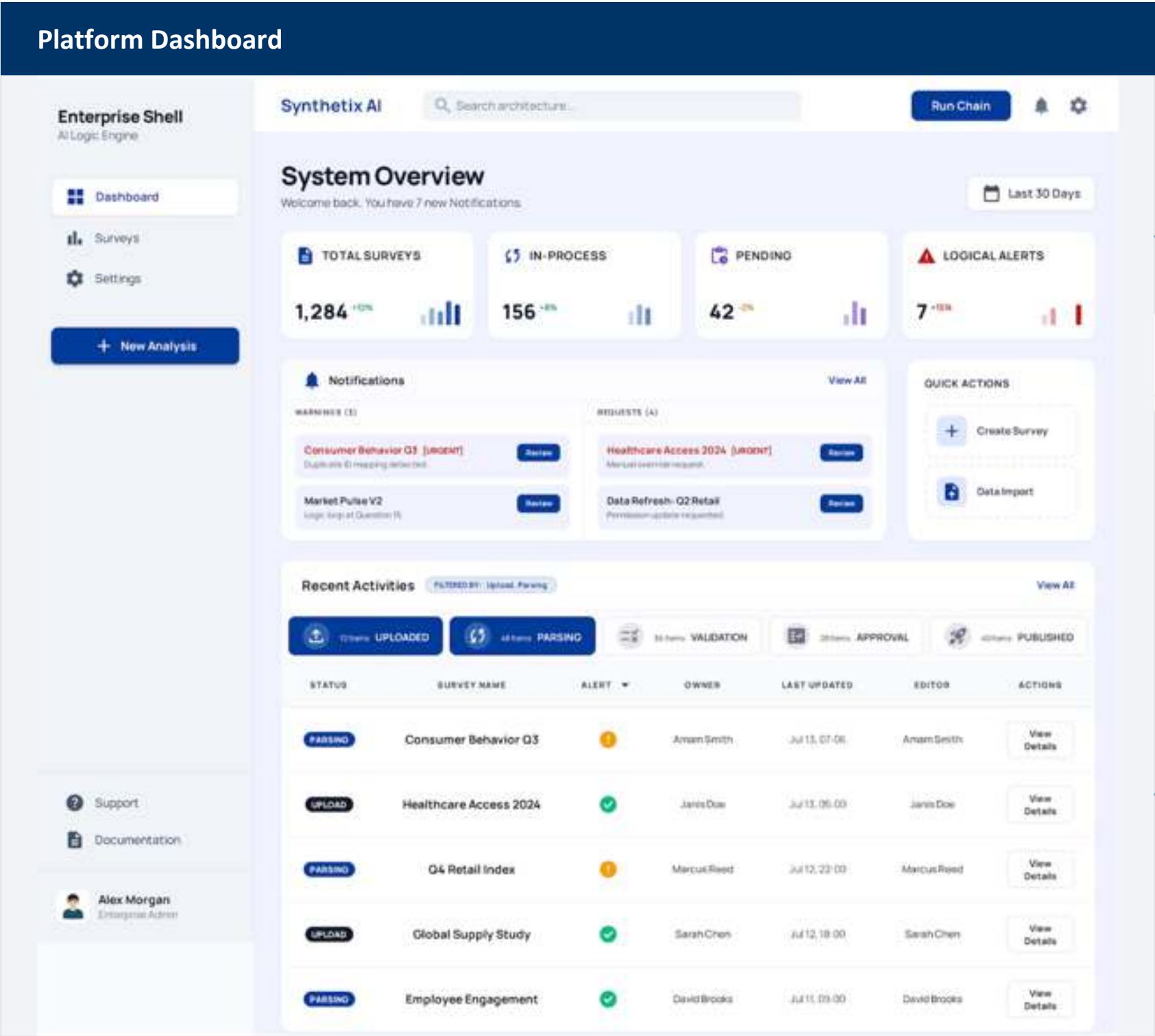
Mechanism	How It Works
Professional Serial Nodes	All stages follow a fixed professional order, preserving risk-control logic
Localized Parallel Editing	All reviewers access the draft simultaneously; each submits feedback independently — no waits
Precision Impact Re-Review	After a change, only affected reviewers are notified; approved, untouched sections are not re-reviewed



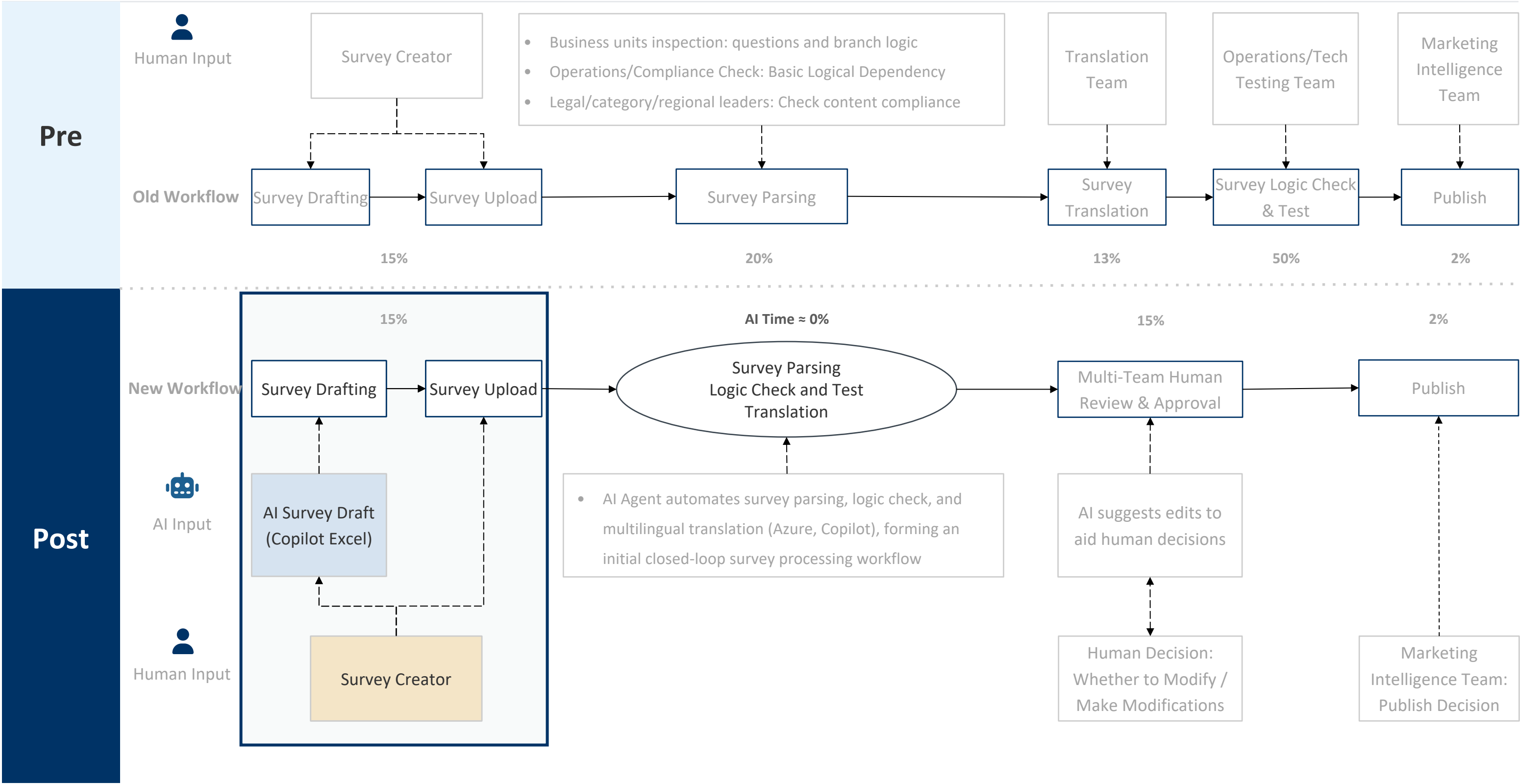
Visualizing the Workflow Transformation



One Platform for All Workflows



Visualizing the Workflow Transformation



One Platform for All Workflows

Survey Drafting Strategy

Strategy:

- Remaining traditional way to prevent accidents and avoid efficiency decrease due to the workflow change.
- Copilot Interaction: By answering structured questions, marketers can efficiently propose a new survey.

Human Roles:

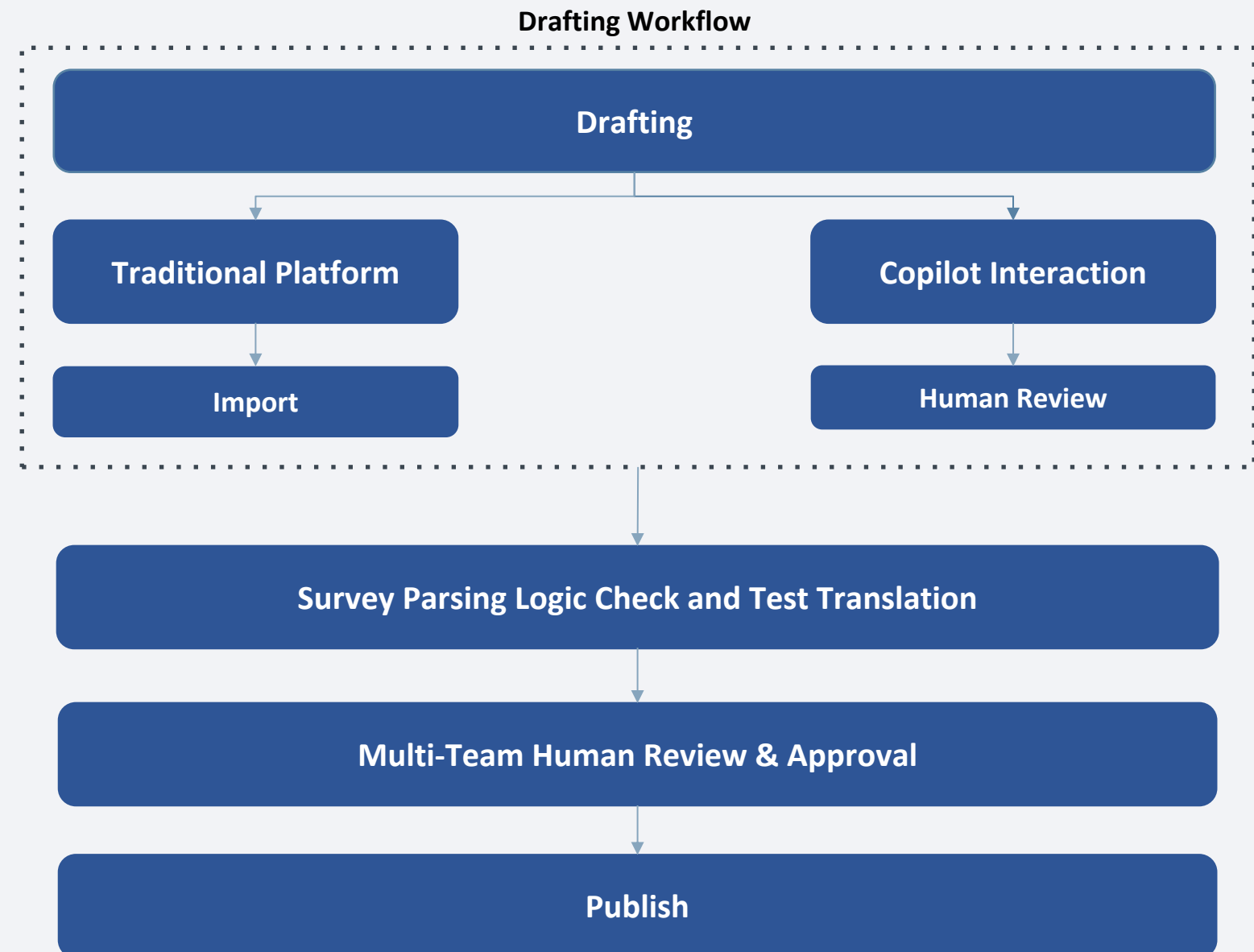
- Define the objective of marketing research;
- Check and modify AI quick results.

Copilot Roles:

- Provide structured questions to help decide research type;
- LLM to utilize historical surveys.

Advantage:

- Exert both the creativity of human and machine-learning ability of AI;
- Avoid always 0-1 process to save time



An integrated and unified control interface for AI execution and human oversight.

One Platform for All Workflows

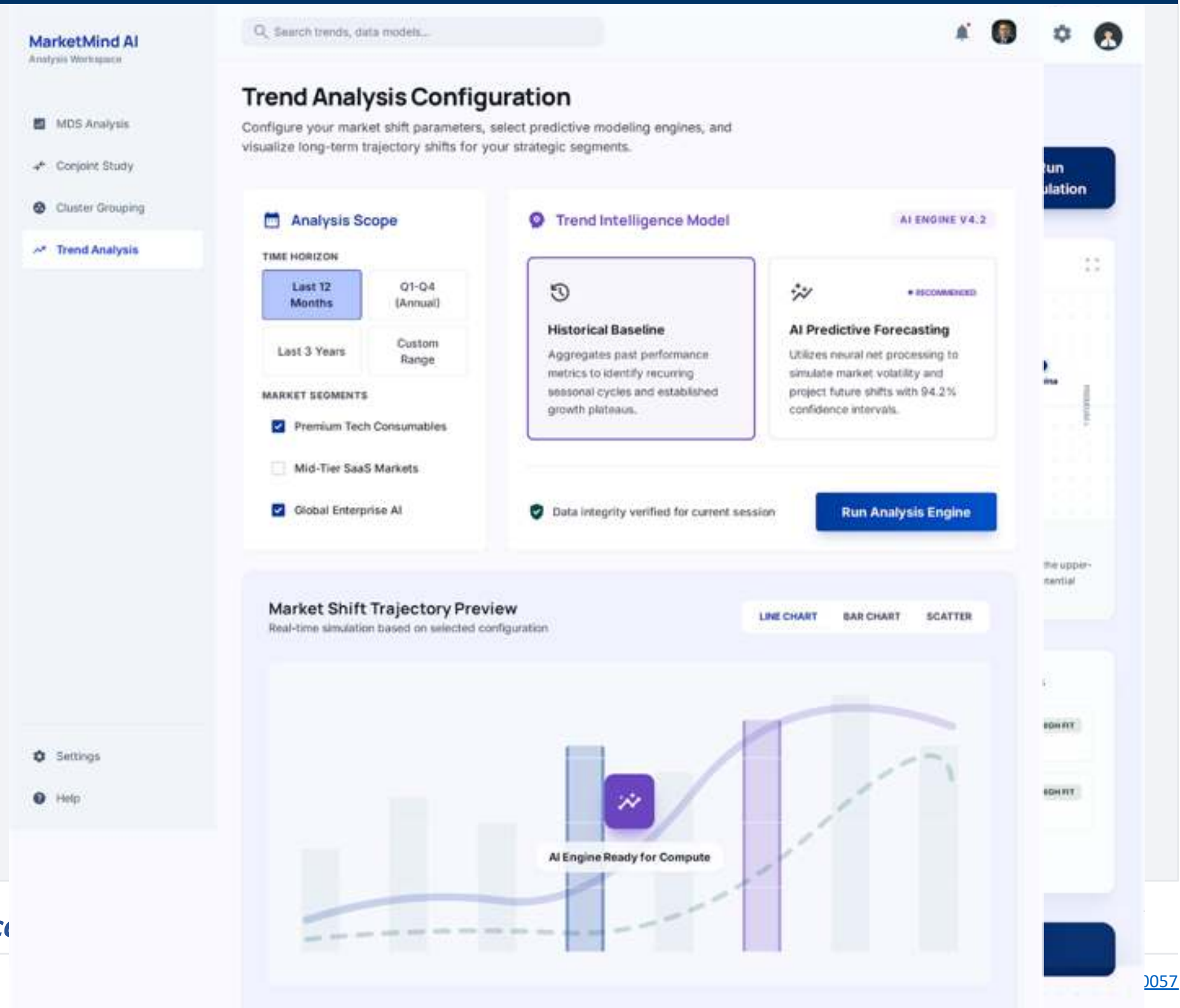
Survey Drafting Function Demonstration

Draft Space:

This workspace aims to improve the efficiency of constructing survey idea.

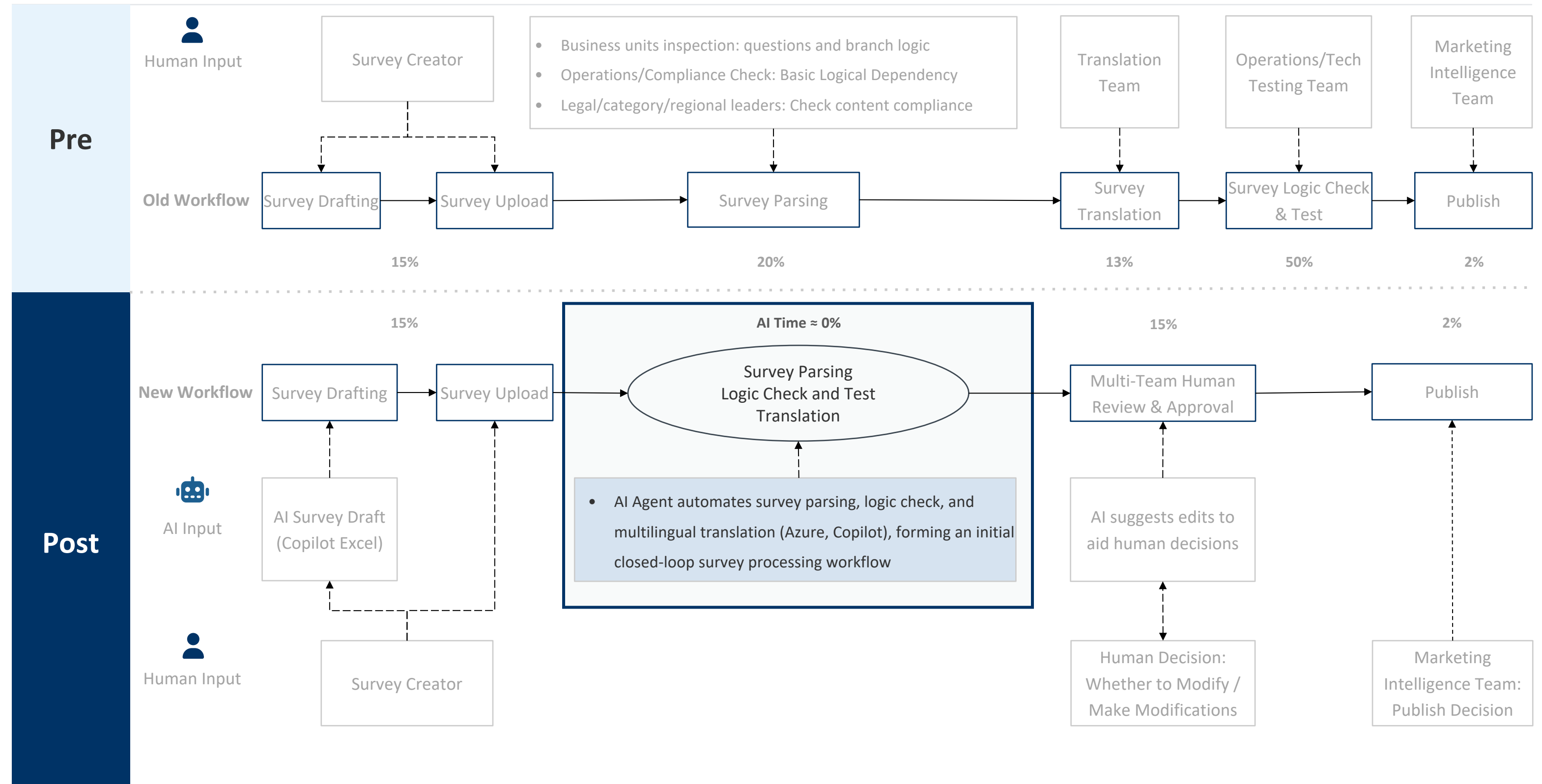
Main Functions :

1. Guided Research Design Tool:
 - a. Define Research Type
 - b. Choose Research Methods
 - c. Historical Templates Apply
2. Platform Export Hub
 - a. Provide all platforms' formats
 - b. Import Format Function
3. 🌟 AI Drafting Workspace
 - a. Creating questions based on specific analysis
 - b. Cluster Analysis
 - c. MDS Analysis
 - d. Conjoint Analysis
 - e. Trend Analysis



An integrated and unified c

Visualizing the Workflow Transformation



Capabilities of the AI Pre-Processing Engine

01

Intelligent Parsing
(Tech: M365 Copilot + Azure OpenAI)

Excel Upload → Logic Tree Visualization

- Parse structured questionnaire data from Excel master template
- Extract skip logic, branching conditions, randomization rules
- Convert to platform-importable format (LSS/JSON)
- Visual logic tree representation for human verification

02

Exhaustive Logic Testing
(Tech: Azure OpenAI + Microsoft Foundry)

5,000+ Path Simulation → Anomaly Report

- Simulate all possible respondent journeys through branching logic
- Detect dead loops, unreachable questions, logic contradictions
- Generate comprehensive test coverage report
- AI-suggested fixes with confidence scores

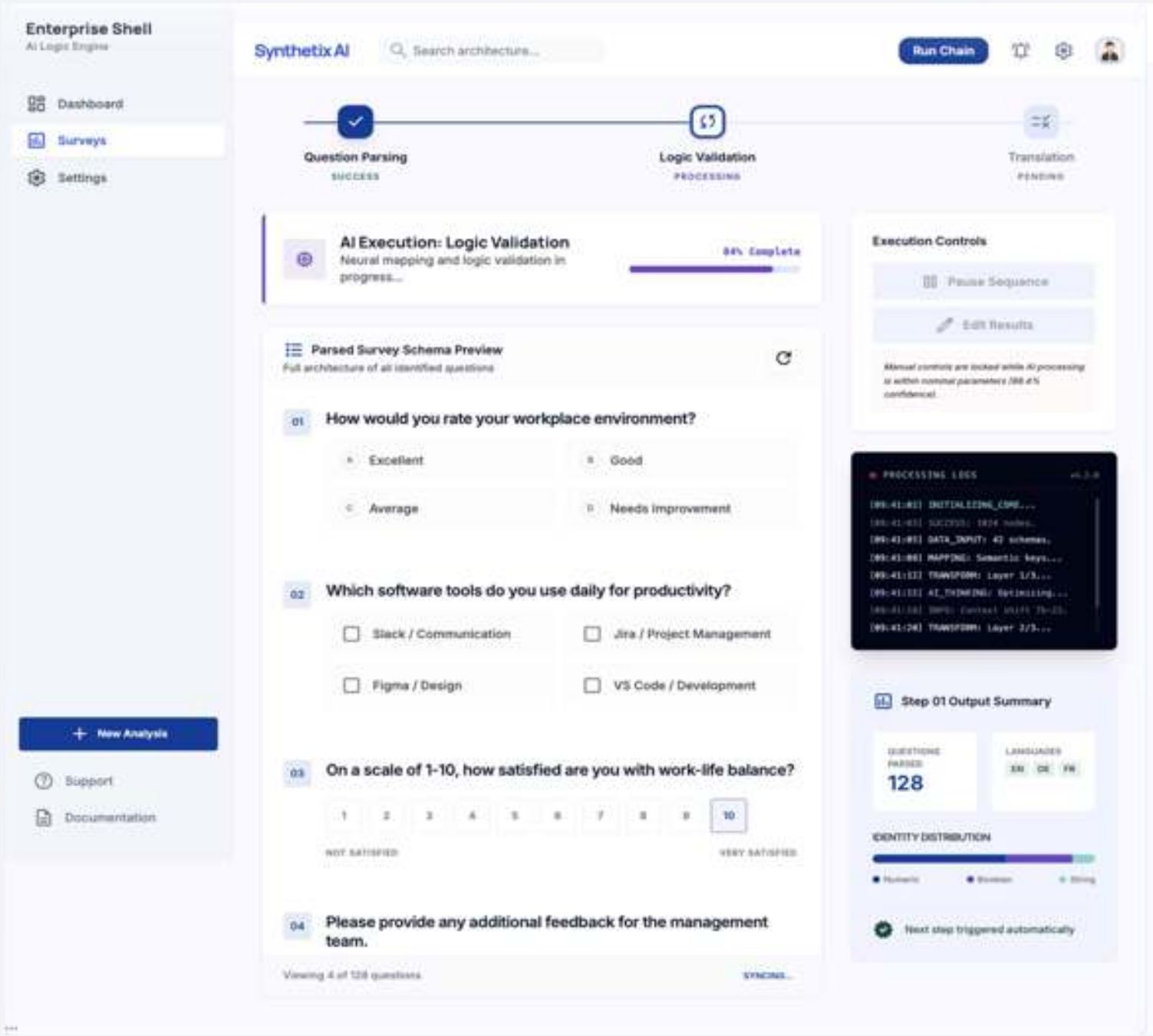
03

Batch Translation & Alignment
(Tech: Azure AI Translator / Azure OpenAI)

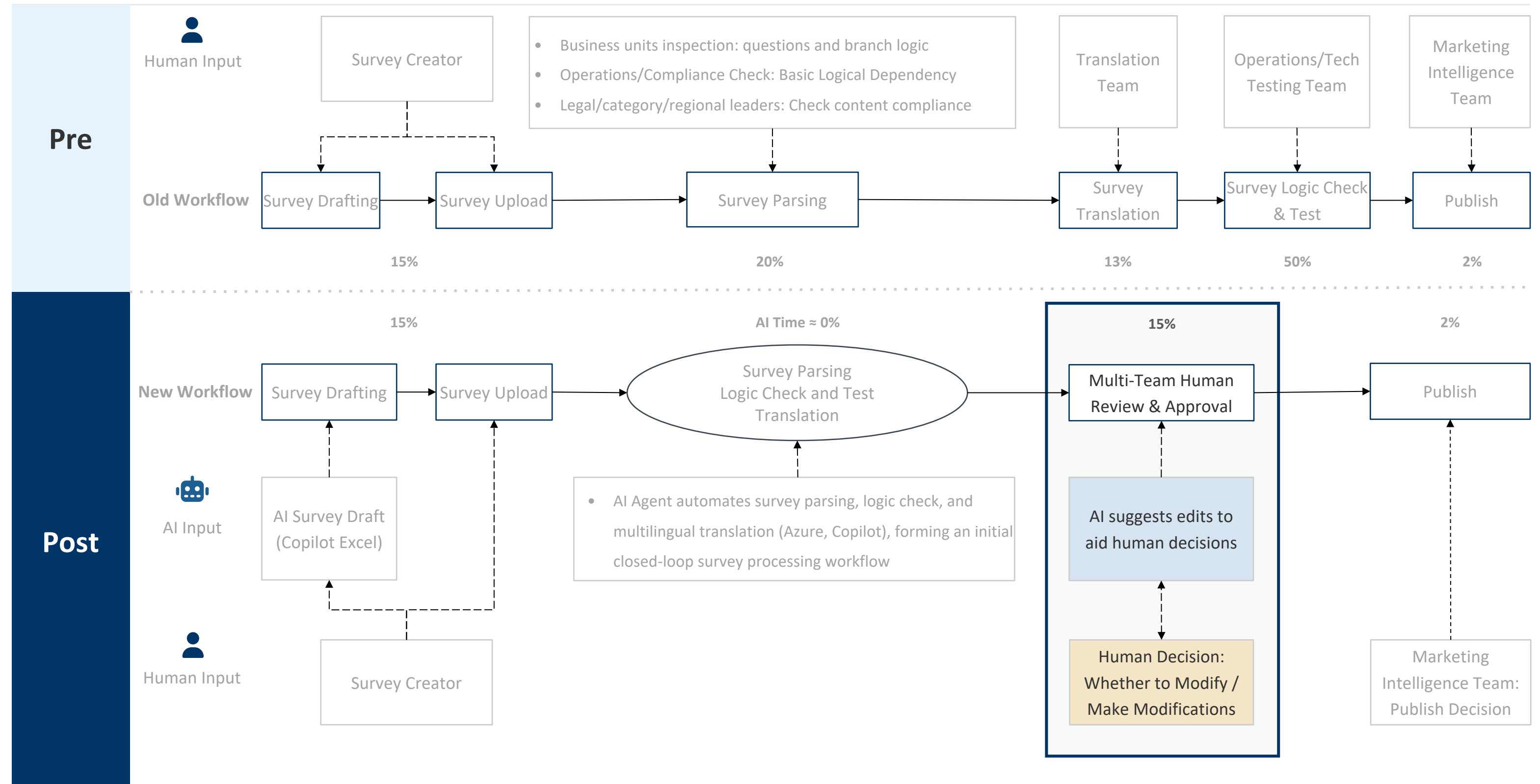
Multi-Language Sync with Terminology Locking

- Batch translate to all 12 required language versions
- Terminology database ensures brand-consistent translation
- Structural alignment validation across all languages
- Cultural sensitivity pre-check with flagging system

Platform Interface Demo



Visualizing the Workflow Transformation



Integrated Review Workbench

Approval Dashboard

InsightWorkbench

BACK TO ALL QUESTIONNAIRES

Q4 Customer Feedback Loop

v2.3

MarketingSubmitted Oct 1140 Questions

01

QUESTION 01

How was your in-person experience during that visit?

Very SatisfiedSatisfiedNeutral

02

QUESTION 02

Rate the speed of our website responsiveness.

Very SatisfiedSatisfiedNeutral

03

QUESTION 03

Which feature do you use most often?

Very SatisfiedSatisfiedNeutral

04

QUESTION 04

Do you find our documentation clear and concise?

Very SatisfiedSatisfiedNeutral

05

QUESTION 05

Terms of use for beta features are acceptable.

Very SatisfiedSatisfiedNeutral

06

QUESTION 06

Very SatisfiedSatisfiedNeutral

07

QUESTION 07

Very SatisfiedSatisfiedNeutral

08

QUESTION 08

Very SatisfiedSatisfiedNeutral

09

QUESTION 09

Very SatisfiedSatisfiedNeutral

10

QUESTION 10

Very SatisfiedSatisfiedNeutral

11

QUESTION 11

Very SatisfiedSatisfiedNeutral

12

QUESTION 12

Very SatisfiedSatisfiedNeutral

13

QUESTION 13

Very SatisfiedSatisfiedNeutral

14

QUESTION 14

Very SatisfiedSatisfiedNeutral

15

QUESTION 15

Very SatisfiedSatisfiedNeutral

16

QUESTION 16

Very SatisfiedSatisfiedNeutral

17

QUESTION 17

Very SatisfiedSatisfiedNeutral

18

QUESTION 18

Very SatisfiedSatisfiedNeutral

Review Progress

MR

SARAH J.

Current

LEGAL

Search Writing

COMPLIANCE

Marissa V.

CATEGORY

David Spivey

REGIONAL

Parthiv

Review

Q22: Missing opt-in statement (GDPR/CCPA)

Q25: Lacks mandatory disclaimer for financial data

Q23: Broken Terms of Service documentation link

View Compliance Editor

Questionnaire Approved Successfully

Integrate All Roles		
Node	Role	Review Content
Node 1	Marketing Intelligence	Ensure that questions align with business goals.
Node 2	Operations	Verify the basic logic are reasonable and conflict-free.
Node 3	Legal	Review content for legal compliance (consumer law, advertising rules)
	Compliance	Validate data privacy, industry-specific regulations, and internal policy adherence.
	Category	Confirm brand tone, category strategy consistency.
	Regional	Assess culturally sensitive terms, and localization quality.
Node 4	Tech	Detect logic loopholes, boundary condition errors, and platform compatibility issues.
Node 5	Translation	Verify translation accuracy and terminology consistency.

Integrated Review Workbench



Final edit and review authority remains **exclusively with human**.

All reviewers in one place

► Centralize all stakeholders into a single workbench.

- Marketing Intelligence
- Operations
- Legal/Compliance/ Category/Regional
- Tech
- Translation



Human-AI Collaboration



AI



Human

- Automatically **analyzes** survey and generates AI suggestions per question.
- Automatically **highlights** the role- specific AI suggestions and generates a **summary**.

- Manually **edit** each question.

- Humans **decide whether to apply** AI suggestions or not.

- Manually **edit** AI suggestions.

Flexible ways to make changes

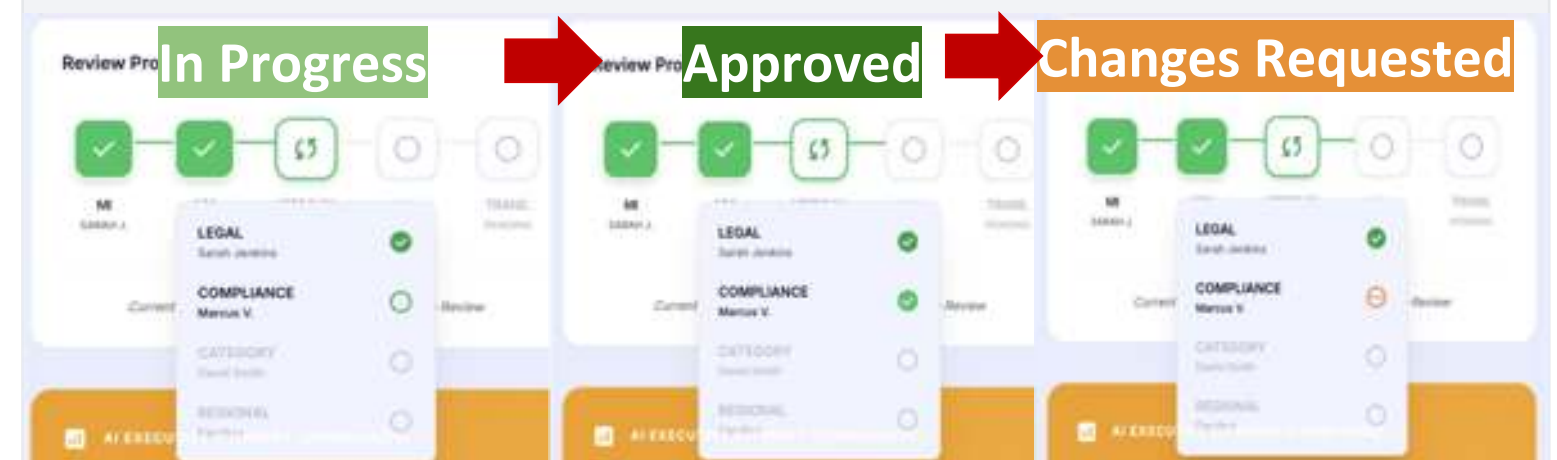
► Directly Change

► Request Change

Related Editors

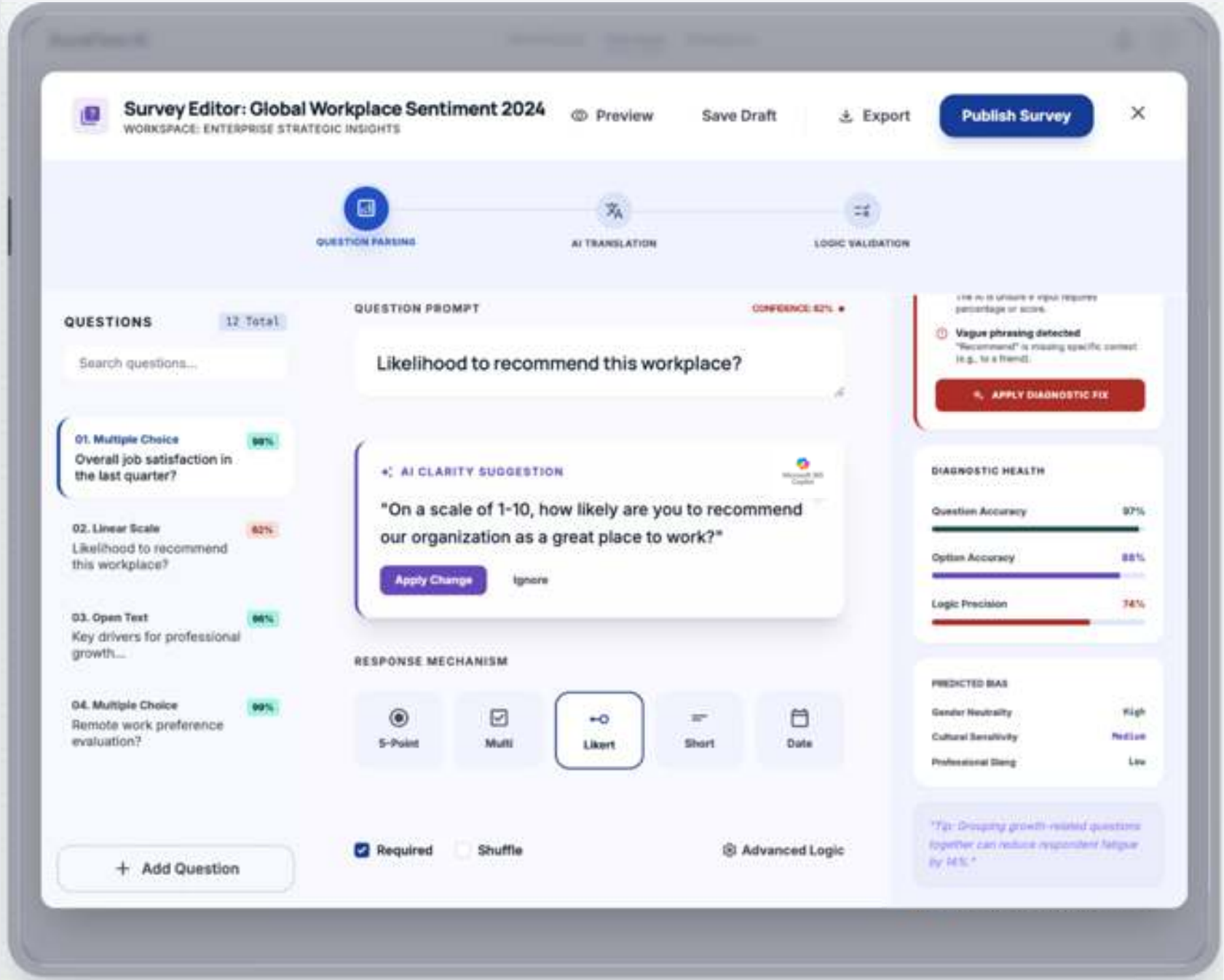
Real-time approval progress

- Display the status of each node clearly, making the entire process **transparent and trackable**.



One Platform for All Workflows

Question Parsing



When there are **relevant comments in the review area** that require manual modification, this page will be redirected.

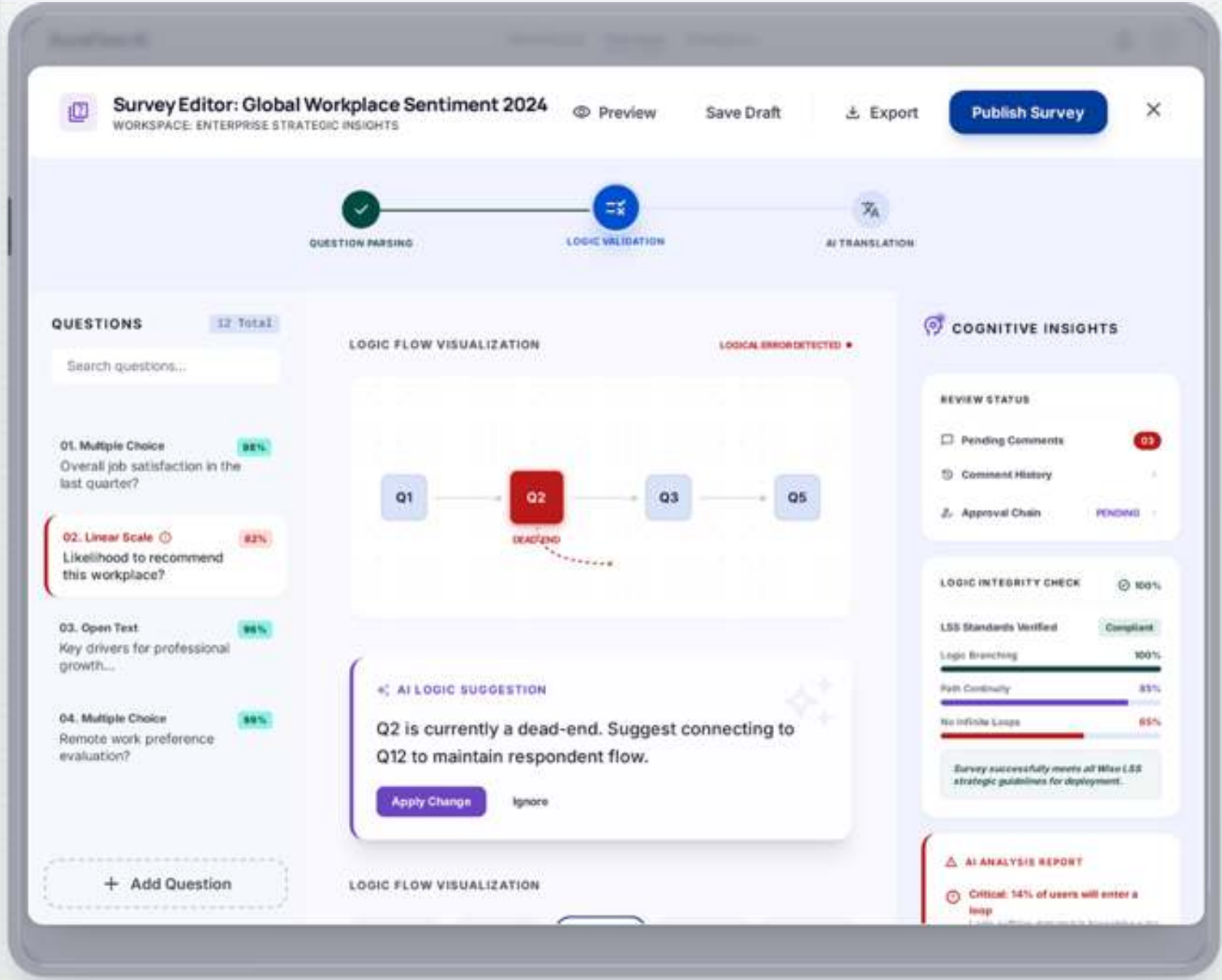
Functional Area

- Review comment area**
Each approval role (legal, category, region) can directly make annotations and @ the responsible person in the comment section when checking corresponding issues.
- Real-time format detection on the Wise platform**
The system automatically generates a Wise LSS file. The page displays "Wise Import Preview" area to check the format compliance for Wise platform.
- AI parsing confidence level & AI summary report**
Display the parsing quality report (accuracy rate of questions, options) for manual review.

An integrated and unified control interface for AI execution and human oversight.

One Platform for All Workflows

Logic Check



When the survey content editing is finalized, the system automatically redirects to this page for a structural integrity audit.

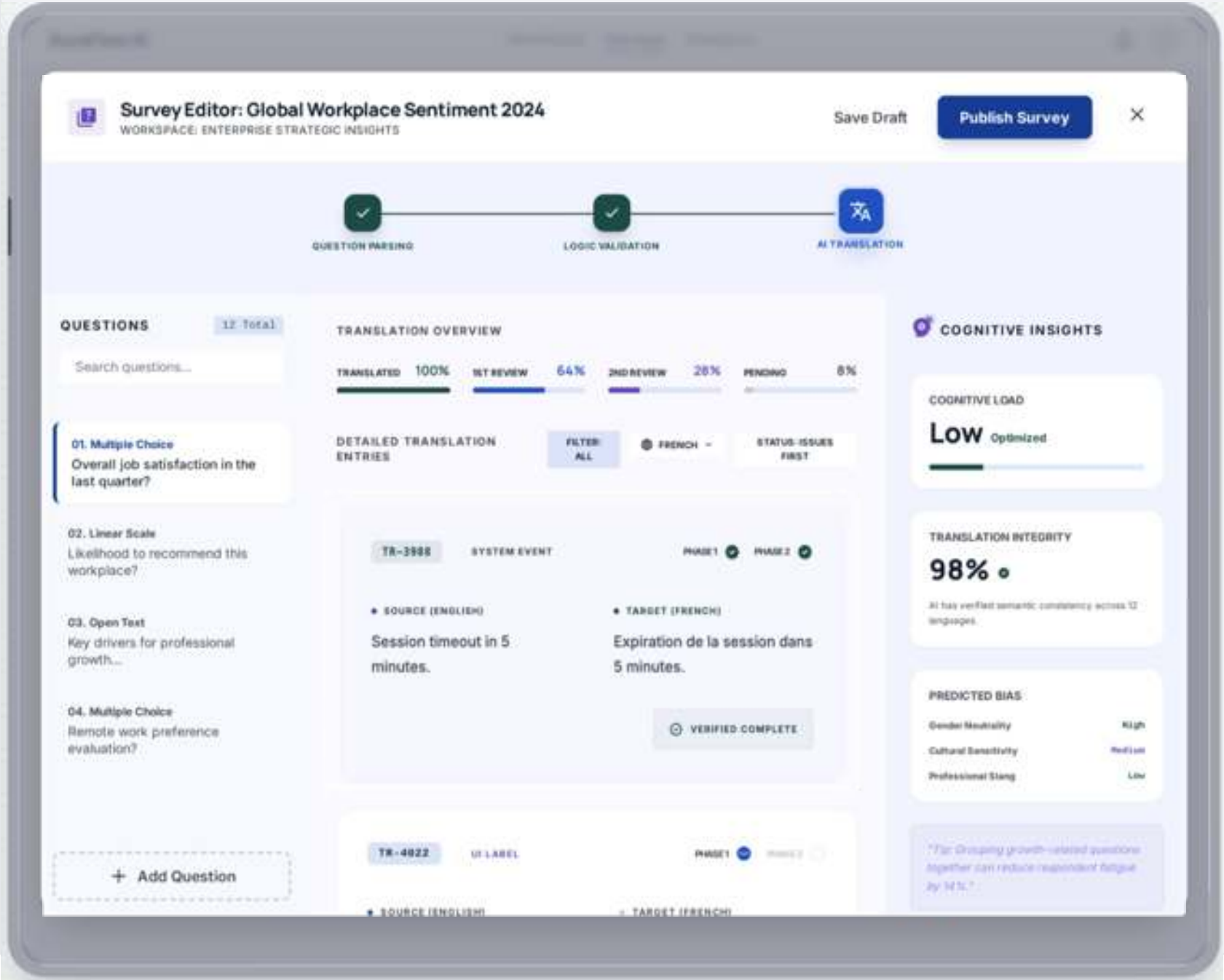
Functional Area

- Interactive Logic Flow Visualization**
The system maps the entire survey architecture into an interactive node-link diagram. Critical logic issues such as dead-ends or infinite loops are highlighted in red, allowing users to visualize the respondent's journey.
- Automated Anomaly Detection & AI Remediation**
AI identifies structural flaws and provides "AI Recommendations." Users can directly interact with these suggestions by selecting Accept to automatically re-route paths or Ignore to maintain manual control.
- Real-time Wise LSS Compliance Audit**
The system performs a specialized check against Wise LSS Strategic Guidelines. It verifies that all branching, piping, and jumping rules are 100% compatible with the Wise platform's import requirements to ensure zero deployment failure.

An integrated and unified control interface for AI execution and human oversight.

One Platform for All Workflows

Questionnaire Translation



COGNITIVE INSIGHTS

1 COGNITIVE LOAD

This indicator aggregates factors such as average sentence length, the proportion of rare or uncommon words, and sentence nesting complexity, in order to assess the readability of the translated questionnaire and minimize comprehension discrepancies.

2 TRANSLATION INTEGRITY

In this example, a completeness score of 98% is displayed, backed by AI verification, which quantitatively measures semantic consistency.

3 PREDICTED BIAS

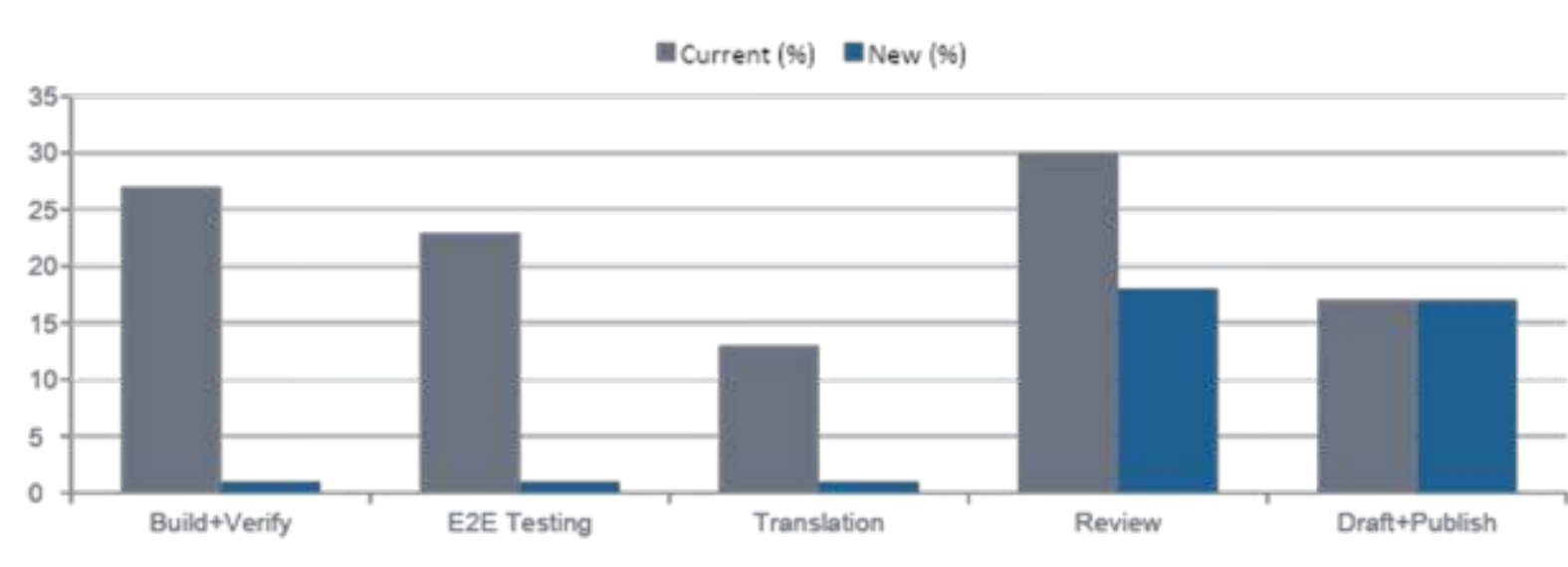
This is further broken down into three dimensions: Gender Neutrality, Cultural Sensitivity, and Technical Terminology. This ensures fairness and validity in cross-cultural research by providing early warnings against potential bias.

An integrated and unified control interface for AI execution and human oversight.

Estimated Operational Impact - KPI (Based on AS Watson Time Data)

Cycle Time Comparison by Workstream

Workstream	Current Time %	New Time %	Reduction	Driver
Build + Verification	27%	<1%	>99%	AI auto-build + verify
End-to-End Testing	23%	<1%	>99%	AI 5,000+ path testing
Translation + Review	13%	<1%	>95%	AI batch translation
Multi-Dept Review	Serial 30%	Integrated 15-20%	~40%	Integrated review model
Drafting + Publication	17%	~17%	Minimal	Human-led (unchanged)
TOTAL CYCLE TIME	100%	~32%	~68% REDUCTION	AI Execution + Integrated Review



~68%

Cycle Time Reduction

Target aligns with AS Watson's stated goal of **50-70% reduction**. The projection is based on:

- AS Watson process time data (10-step breakdown)
- Mechanism redesign (AI automation + parallel review)
- Industry benchmarks for similar AI implementations

Note: Projections based on AS Watson process data and mechanism redesign. Actual results depend on implementation scope and organizational adoption.

Comparable Success in the Field

FINANCIAL SERVICES

Avannis

GenAI survey coding automation saved

3,000 work hours/year

Context: Monthly manual coding of 20,000+ customer surveys. Labor-intensive categorization and sentiment analysis process.

Solution: Implemented Amazon Bedrock (Anthropic Claude) GenAI solution for automatic survey classification and coding.

Key Takeaway: **Scalability without headcount increase** — handling growing survey volumes without adding staff.

"GenAI solution expected to save 3,000 work hours annually." — CTO, Avannis

SaaS PLATFORM

Drag'n Survey

AI Agent reduced survey creation time to

1/10th of original

Context: French survey platform expanding to India market. Need to simplify survey creation for non-technical users.

Solution: Launched multimodal AI Agent — users describe needs in natural language, AI builds complete survey and analyzes results.

Key Takeaway: **Natural language to structured output** — democratizing survey creation without technical expertise.

Launched in India with AI Agent to transform online surveys.



Uncertainty Control Note: Case studies validate the feasibility of AI-assisted survey workflows while emphasizing human oversight remains essential. Both implementations maintain human review gates — AI accelerates execution, but humans retain decision authority.

Sources: Avannis case study — Innovative Solutions (innovativesol.com); Drag'n Survey India launch — ANI News ([aninews.in](https://ani.news))

The Goal is Not “AI Everywhere”, It’s “The Right Role”



Efficiency Source

AI does the heavy lifting (Execution):

- Intelligent parsing of Excel templates
- Exhaustive logic path testing (5,000+)
- Batch translation to all languages
- Automated test case generation

Humans eliminate wait times (Coordination):

- Parallel review across all departments
- Simultaneous access, no serial blocking



Risk Mitigation

Precision Impact Re-Review:

- Smart tagging auto-labels questions by department
- Only affected parties are notified of changes
- Eliminates unnecessary re-review cycles

Immutable modification records:

- Every edit timestamped, attributed, and preserved
- Complete before/after diff for every change

Human Gateway for Publish:

- Final sign-off remains exclusively human



Scalability

Designed once, deployable everywhere:

- All AS Watson markets and survey types
- Same architecture, different content
- 5-10+ languages handled simultaneously

Cross-use case migration:

- "AI Pre-Proc + Integrated Review" model extends to other use cases
- Same governance, same traceability

”

Fast without accuracy is useless. Accurate without speed doesn't scale.
We aim for Fast and Accurate.

Our proposal defines clear boundaries:

AI executes repetitive tasks with speed and consistency; humans provide judgment, oversight, and final approval.

Extending the Value

PHASE 1

System Integration

Future API connectivity with Wise platform for full end-to-end automation.

Currently, the Wise survey platform is **not API-enabled**.

Once API access is established with the vendor:

- Complete Excel-to-publish automation
- Eliminate all manual platform data entry
- True zero-touch survey deployment

Prerequisite: Vendor API enablement

PHASE 2

Knowledge Transfer

Apply the same architecture to other use case.

The “AI Pre-Processing + Integrated Review” model is domain-agnostic:

- AI pre-checks
- Integrated review by Legal, Brand, Regional teams
- Same precision impact re-review mechanism
- Same immutable audit trail

Same architecture, different domain

PHASE 3

Continuous Learning

AI refinement based on human review patterns and updated brand lexicons.

The system grows smarter over time:

- Terminology databases expand with approved surveys
- Compliance rules evolve from review feedback
- Translation quality improves from human corrections
- Pattern recognition for common issues

Flywheel effect: more usage → better AI

Thank You!

Questions & Discussion



HKU Capstone Group 1

HKU Business School | In collaboration with Microsoft & AS Watson

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